

Department of Computer Science & Engineering, Punjabi University, Patiala

MST-1

MM: 15

B.Tech-3rd Sem (CSE)

Time: 1 hour

Data Structure (CSEB2302T)

Note: Q1 is Compulsory. Attempt any two questions from Q2, Q3 and Q4 each question Carrying five marks.

Q1. Explain the purpose of following (5 Marks)

- a) Time complexity (1 Marks) b) Column-major order 2D Array Memory Management (1 Marks)
c) Two way list working (1 Marks) d) Chaining in Hashing (2 Marks)

Q2. Explain the Concept of Binary search with suitable example.

Q3. How Stack and Queue is implemented with Link List? Explain in detail with their Algorithm for Insertion and Deletion.

Q4. Apply Selection Sort on 77, 33, 44, 11, 88, 22, 66, 55 and write it's Algorithm.

Department of Computer Science & Engg.

Sem: 3rd

Sub: Data Structures

Max Marks: 15

Time: 60 Minutes

Section-A (Attempt All Questions)

1. Why we have different types of data Structures? (1)
2. Write any two limitations of arrays. (2)
3. List the various operations that can be performed on queues. (2)

Section-B (Attempt Any Two Questions) 5 Marks Each

1. Write an algorithm to search an element in an array using binary search technique.
2. Write an algorithm to sort given elements using bubble sort technique:
55, 33, 77, 44, 66, 88, 11, 22
3. How stacks are used while performing recursion? Take suitable example.

SUBJECT: DATA STRUCTURE (MST-1)

DATE: 14-OCT-2021 Class: BTECH 2nd yr (3rd SEM)

MM: 15

EXAMINER: Ms. Harpreet Kaur & Ms. Anantdeep Kaur

Section A (5*1=5)

Q: Calculate the complexity of Bubble sort for best case and worst case.

Q: Write one application for each Circular Queue and Stack.

Q: What are the merits and demerits of a linear queue over circular queue?

Q: What is the need of Linked List and why we use pointers in linked list, write any application?

Q: Write an algorithm to for Binary Search.

Section B (3*5=15) (Q2 is compulsory)

Q2:a) Write output of following C code (Give Explanation):

(2 marks)

Suppose you are given an array $s[1 \dots n]$, and a procedure $\text{reverse}(s, i, j)$ which reverses the order of elements in between positions and j (both inclusive). What does the following sequence do, where?

```

1 <= k <= n:
reverse(s, 1, k);
reverse(s, k + 1, n);
reverse(s, 1, n);

```

1 2 3 4 5
5 4 3 2 1
5 4 1 2 3
3 2 1 4 5

i) Rotates s by k positions ii) Leaves s unchanged iii) Reverses all the elements of s iv) None

b) Following code modifies the given singly linked list by moving the last element to the front of the list and returns the modified list. Some part of the code is left blank. (Give Explanation)

```

typedef struct node
{
    int value;
    struct node *next;
} Node;
Node *move_to_front(Node *head)
{
    Node *p, *q;
    if (head == NULL) || (head->next == NULL)
        return head;
    q = NULL;
    p = head;
    while (p->next != NULL)
    {
        q = p;
        p = p->next;
    }
    return head;
}

```

Choose the correct alternative to replace the blank line.

- A. $q = \text{NULL}; p \rightarrow \text{next} = \text{head}; \text{head} = p;$
B. $q \rightarrow \text{next} = \text{NULL}; \text{head} = p; p \rightarrow \text{next} = \text{head};$
C. $\text{head} = p; p \rightarrow \text{next} = q; q \rightarrow \text{next} = \text{NULL};$
D. $q \rightarrow \text{next} = \text{NULL}; p \rightarrow \text{next} = \text{head}; \text{head} = p;$

Q3: a) Write an algorithm to delete first and last node of linked list.

(3)

b) Write an algorithm to print last element of Linked List.

(2)

OR

Q4: a) Write Algorithm of Quick sort with an example.

(3)

b) Also calculate complexity of Quick Sort.

(2)

Department of Computer Science and Engineering
Punjabi University, Patiala

DATA STRUCTURES (MST - I)

Time Duration: 1 Hr.

Max Marks: 15

Note: All the Questions are compulsory in Section A. Attempt any two questions from Section B

Section-A

- Q1. Differentiate between stacks and queues. (1)
Q2. Write an algorithm to insert an element in an array at the beginning. (2)
Q3. How stacks are used in recursion while finding the factorial of 5. (2)

Section-B

- Q4. Write an algorithm of bubble sort. Count the number of passes and comparisons in the given list: 56, 23, 10, 88, 44, 5. (2x5=10)
Q5. Write an algorithm for binary search using a suitable example.
Q6. Write an algorithm to insert an element 44 at 3rd position in a given linked list: 11, 22, 33, 55.

Department of Computer Science & Engineering, Punjabi University, Patiala

MST-1

B.Tech-3rd Sem (CSE)

MM: 15

Time: 1 hour

Data Structure

Note: Q1 is Compulsory. Attempt any two questions from Q2, Q3 and Q4, each question carrying 5 marks.

- | | | |
|------------|---|---|
| Q1. | a) What do you mean by Time-Space Complexity | 1 |
| | b) What do you understand by Hashing? | 1 |
| | c) How did Stack work? Explain with a diagram. | 1 |
| | d) Apply Merge sort on 78, 20, 33, 56, 10, 65 | 2 |
| Q2. | Write the algorithm of Binary Search and explain its working with a suitable example | 5 |
| Q3. | Write the algorithms for insertion, deletion and searching in a linked list and explain with suitable diagrams. | 5 |
| Q4. | Apply Radix sort on 425, 110, 566, 235, 785, 911, 665, 157, 999 | 5 |